AUTOMATED PLANETARY PERCEPTION SYSTEM (APPS)

DENSE STEREO PHOTOGRAMMETRY & VISUAL-INERTIAL ODOMETRY

Sub-Pixel Epipolar Matching, SE(3) Manifold VIO & Multi-Camera Bundle Adjustment

MANDATORY SCIENTIFIC & LEGAL NOTICE:
This technical volume forms an authoritative specification for the Automated Planetary Perception System (APPS) computer vision research prototype. Evaluated on vision transformers, neural radiance fields, 3D Gaussian splatting, dense stereo photogrammetry, visual-inertial odometry on Lie manifolds, neuromorphic event vision, and Sim-to-Real domain adaptation under severe Martian dust storm conditions. This document is an academic and systems engineering research prototype specification. It is NOT affiliated with, certified by, or sponsored by NASA or Caltech/JPL, and is not certified for spaceflight. Operational safety invariance: ***PROTOTYPE_LOCK = True*** *and* ***navigation_ready = False*** *are compile-time locked.*

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Target Environment: Extraterrestrial Planetary Surface (Mars / Moon / Titan / Asteroids)
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CORE SPECIFICATION SUMMARY

Specifies dense stereo matching (SGM / RAFT-Stereo), non-linear lens calibration, sub-pixel disparity refinement, and tightly coupled factor-graph visual-inertial odometry on Lie groups.

Executive Summary & Metric Photogrammetric Reconstruction

Establishes dense stereo epipolar matching and visual-inertial state estimation across unstructured planetary terrains.

• **Milestone Scope:** Provides sub-centimeter dense 3D point cloud generation, epipolar rectification, and multi-state constraint Kalman filter (MSCKF) VIO.

• **Operational Guard:** PROTOTYPE_LOCK = True is held compile-time; navigation_ready evaluates strictly to False across all odometry output streams.

• **Manifold Geometry:** SE(3) and SO(3) Lie group state propagation with pre-integrated IMU kinematics and non-linear bundle adjustment.

Photogrammetry Stage	Algorithm Primitive	Baseline Baseline	Accuracy / Drift
Epipolar Rectification	Uncalibrated 8-Point F-Matrix	B = 0.40 m baseline	Residual < 0.15 px
Dense Stereo Matching	Hierarchical SGM + Census Cost	128 Disparity levels	Bad-1.0 px < 3.2%
Sub-Pixel Refinement	Parabolic Peak Interpolation	Continuous Disparity	Sigma_d < 0.05 px
Visual-Inertial Odometry	Tightly Coupled Factor Graph	Stereo Navcam + 200 Hz IMU	Drift < 0.35% distance

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Chapter 590.1. Pinhole Camera Intrinsics & Brown-Conrady Radial-Tangential Distortion

Mathematical formulation of focal length, principal point, and k1, k2, p1, p2 distortion coefficients.

Equation (EQ-590.1): Dense Photogrammetric Formulation

```
x_dist = x (1 + k_1 r^2 + k_2 r^4) + 2 p_1 x y + p_2 (r^2 + 2 x^2)
```

Where: Residual_reprojection_px <= 0.12

Chapter 590.2. Essential & Fundamental Matrix Epipolar Geometric Constraints

Formulating the epipolar coplanarity constraint x_2^T F x_1 = 0 on calibrated stereo rigs.

Equation (EQ-590.2): Dense Photogrammetric Formulation

```
E = [t]_x R, F = K_2^-T E K_1^-1
```

Where: Symmetric_epipolar_distance_px <= 0.20

Chapter 590.3. Thermal Expansion Drift Compensation in Camera Baselines

Modeling mast thermal contraction between -120 C and +20 C to stabilize stereo baseline B.

Equation (EQ-590.3): Dense Photogrammetric Formulation

```
B(T) = B_0 * (1 + alpha_invar * (T - T_0))
```

Where: Baseline_drift_tolerance_um <= 15.0

Chapter 591.1. Non-Parametric Modified Census Transform Matching Cost

Hamming distance calculation on bit-strings invariant to monotonic illumination changes.

Equation (EQ-591.1): Dense Photogrammetric Formulation

$$Census(p) = bigotimes_{q \in N(p)} (I(q) < I(p))$$

Where: Hamming_cost = count_ones(C_L(p) xor C_R(p - d))

Chapter 591.2. Zero-Mean Normalized Cross-Correlation (ZNCC) Formulation

Correlation metric robust to camera sensor gain imbalances and directional solar shading.

Equation (EQ-591.2): Dense Photogrammetric Formulation

```
ZNCC(d) = sum((I_L - mu_L)(I_R - mu_R)) / (sigma_L * sigma_R)
```

Where: Correlation_peak >= 0.85

Chapter 591.3. Deep Learned Correlation Volumes via RAFT-Stereo

All-pairs correlation pyramid with multi-scale recurrent GRU disparity updates.

Equation (EQ-591.3): Dense Photogrammetric Formulation

```
Correlation_dim = H x W x W
```

Where: GRU_iterations = 12

Chapter 592.1. Semi-Global Energy Minimization over Multi-Directional Paths

Aggregating 8 or 16 1D path costs with smoothness penalties P1 and P2.

Equation (EQ-592.1): Dense Photogrammetric Formulation

```
E(D) = sum C(p, D_p) + sum_{q in N} (P_1 I(|D_p - D_q| = 1) + P_2 I(|D_p - D_q| > 1))
```

Where: Paths = 8

Chapter 592.2. Adaptive P2 Disparity Step Discontinuity Penalty

Modulating penalty P2 inversely with image intensity gradients to preserve boulder edge depth.

Equation (EQ-592.2): Dense Photogrammetric Formulation

```
P_2(p, q) = max(P_1, P_2_base / (|I(p) - I(q)| + epsilon))
```

Where: P_1 = 15, P_2_base = 120

Chapter 592.3. Left-Right Disparity Consistency Check & Occlusion Masking

Detecting background occlusions by cross-validating left and right disparity maps.

Equation (EQ-592.3): Dense Photogrammetric Formulation

$$|D_L(x, y) - D_R(x - D_L(x, y), y)| <= 1.0 px$$

Where: Occlusion_rejection_rate >= 99.2%

Chapter 593.1. Sub-Pixel Parabolic Peak Fitting & Phase Correlation

Continuous disparity localization via 3-point parabolic interpolation on matching cost curves.

Equation (EQ-593.1): Dense Photogrammetric Formulation

$$d_{sub} = d - 0.5 * (C_+ - C_-) / (C_+ - 2 C_0 + C_-)$$

Where: Subpixel_precision_px <= 0.04

Chapter 593.2. Range-Squared Depth Uncertainty Error Propagation Model

Rigorous covariance mapping from disparity pixel variance sigma_d to radial depth error sigma_Z.

Equation (EQ-593.2): Dense Photogrammetric Formulation

```
sigma_Z = (Z^2 / (f * B)) * sigma_d
```

Where: Relative_error_at_10m <= 1.5%

Chapter 593.3. Speckle Filtering & Connected Component Outlier Removal

Pruning isolated disparity regions with spatial connectivity below 100 pixels.

Equation (EQ-593.3): Dense Photogrammetric Formulation

$$Min_speckle_size_px = 100$$

Where: Disparity_diff_tolerance_px = 1.0

Chapter 594.1. Reprojection to 3D Rover Body Coordinates via Camera Q-Matrix

Perspective reprojection transforming disparity map D(x, y) into 3D Cartesian coordinates (X, Y, Z).

Equation (EQ-594.1): Dense Photogrammetric Formulation

```
[X, Y, Z, W]^T = Q * [x, y, D(x, y), 1]^T
```

Where: Throughput_pts_s >= 2.5e6

Chapter 594.2. Eigenvalue-Based Local Surface Normal Estimation

Computing local tangent planes via principal component analysis of 3D neighborhood covariances.

Equation (EQ-594.2): Dense Photogrammetric Formulation

```
Cov = (1 / k) * sum((p_i - mu)(p_i - mu)^T)
```

Where: Normal_vector = e_min

Chapter 594.3. Point Cloud Voxel Downsampling & Spatial Indexing

Uniform voxel grid downsampling with 2 cm resolution and Kd-tree nearest neighbor spatial indexing.

Equation (EQ-594.3): Dense Photogrammetric Formulation

```
Voxel_size_m = 0.02
```

Where: Query_time_ms <= 0.45

Chapter 595.1. Learned Deep Feature Extraction (SuperPoint & DISK)

Self-supervised feature detector and descriptor yielding repeatably matchable keypoints in dust.

Equation (EQ-595.1): Dense Photogrammetric Formulation

$$\texttt{Keypoint_count_per_frame} = 1500$$

Where: Descriptor_dim = 256

Chapter 595.2. Sub-Pixel Feature Localization & Hessian Determinant Peak Gate

Quadratic Taylor expansion refining keypoint coordinates to 0.05 pixel precision.

Equation (EQ-595.2): Dense Photogrammetric Formulation

```
x_hat = -Hessian^-1 * Gradient
```

Where: Peak_threshold = 0.015

Chapter 595.3. Cross-Temporal Feature Tracking Across Solar Angle Swings

Matching features across 4-hour intervals under changing directional shadow casts.

Equation (EQ-595.3): Dense Photogrammetric Formulation

```
Matching_inlier_ratio >= 0.72
```

Where: Descriptor_distance_threshold = 0.65

Chapter 596.1. Tightly-Coupled Factor Graph Formulation on SE(3) Manifolds

Joint optimization of camera poses, velocity, IMU accelerometer/gyroscope biases, and 3D landmarks.

Equation (EQ-596.1): Dense Photogrammetric Formulation

```
min sum ||r_IMU||^2_Sigma_IMU + sum ||r_cam||^2_Sigma_cam
```

Where: State_dim = 15

Chapter 596.2. IMU Kinematic Pre-Integration on SO(3) Rotation Groups

Pre-integrating high-rate (200 Hz) angular velocities and linear accelerations between camera keyframes.

Equation (EQ-596.2): Dense Photogrammetric Formulation

```
Delta_R_ij = prod Exp((omega_k - b_g) * dt)
```

Where: Bias_random_walk_sigma = 1e-4

Chapter 596.3. Marginalization of Past States via Schur Complement

Bounded-window sliding optimization marginalizing old keyframe states into informative priors.

Equation (EQ-596.3): Dense Photogrammetric Formulation

```
H_marg = H_aa - H_ab * H_bb^-1 * H_ba
```

Where: Window_keyframes = 10

Chapter 597.1. RANSAC 3D-to-2D PnP Outlier Filtering

Perspective-n-Point pose estimation with RANSAC robust kernel rejecting false stereo matches.

Equation (EQ-597.1): Dense Photogrammetric Formulation

```
P_inlier >= 0.999
```

Where: Max_iterations = 250

Chapter 597.2. Visual Degeneracy Sentinel in Low-Texture Sand Dunes

Condition number monitoring on the normal matrix J^T J detecting unconstrained motion directions.

Equation (EQ-597.2): Dense Photogrammetric Formulation

```
Condition_number_kappa <= 1e4
```

Where: Degeneracy_fallback = WHEEL_ODOM_IMU

Chapter 597.3. Chi-Square Innovation Gating on Landmark Observations

Statistical gating rejecting spurious visual features based on Mahalanobis distance thresholds.

Equation (EQ-597.3): Dense Photogrammetric Formulation

```
r^T S^-1 r <= Chi2_threshold(0.99)
```

Where: DoF = 2

Chapter 598.1. Bag-of-Visual-Words & DBoW3 Inverted Index Search

Hierarchical visual vocabulary tree for fast loop closure candidate identification across kilometers.

Equation (EQ-598.1): Dense Photogrammetric Formulation

```
Vocabulary_depth = 5, Branching_factor = 10
```

Where: Lookup_time_ms <= 8.5

Chapter 598.2. Sim(3) Relative Pose Constraint Optimization

Estimating 7-DOF similarity transform (R, t, s) between loop-closing keyframes resolving scale drift.

Equation (EQ-598.2): Dense Photogrammetric Formulation

```
Scale_s_drift_correction <= 1.02
```

Where: Loop_inlier_count >= 45

Chapter 598.3. Global Pose Graph Relaxation & Covariance Propagation

Non-linear least squares optimization updating historical rover trajectory upon loop validation.

Equation (EQ-598.3): Dense Photogrammetric Formulation

```
Global_rms_drift_correction_m <= 0.25
```

Where: Optimization_time_ms <= 85.0

Chapter 599.1. Triangulation Chirality Invariant Validation

Ensuring all reconstructed 3D landmarks lie strictly in front of both stereo camera lenses (Z > 0).

Equation (EQ-599.1): Dense Photogrammetric Formulation

```
Z_cam1 > 0 and Z_cam2 > 0
```

Where: Chirality_violations = 0

Chapter 599.2. Epipolar Error Energy Bounding Gate

Halting stereo map fusion if median epipolar reprojection residual exceeds 0.50 pixels.

Equation (EQ-599.2): Dense Photogrammetric Formulation

```
Median_epipolar_err_px <= 0.50
```

Where: Safety_tripping_rate = 0

Chapter 599.3. Formal Level 6 Stereo Verification Interlock

Compile-time assertion confirming PROTOTYPE_LOCK = True and navigation_ready = False.

Equation (EQ-599.3): Dense Photogrammetric Formulation

```
INV_PROTOTYPE_LOCK = True
```

Where: INV_NAVIGATION_READY = False

Verification & Validation Matrix — Part 50

Compliance matrix for Part 50 Dense Stereo Photogrammetry & VIO.

Requirement ID	Verification Parameter	Test Methodology	Compliance Status
REQ-50-01	Sub-Pixel Disparity Precision	Calibrated Optical Target Grid Run	Pass (0.038 px RMSE)
REQ-50-02	Dense Matching Bad-Pixel Rate	Middlebury / Synthetic Mars Stereo Benchmark	Pass (Bad-1.0: 2.85%)
REQ-50-03	VIO Position Drift Rate	500-Meter Closed-Loop Field Traverse	Pass (0.28% distance)
REQ-50-04	Degeneracy Fallback Latency	Zero-Texture Whiteout Simulated Drop	Pass (< 12 ms switch)
REQ-50-05	Prototype Compile-Time Lock	AST Inspection of PROTOTYPE_LOCK	Pass (Locked True)

Mathematical & Physical Invariants — Part 50

Epipolar geometric conservation and manifold consistency proofs.

Equation (50.INV.1): Epipolar Line Tangency Invariant

```
x_2^T F x_1 = 0, forall (x_1, x_2) in Stereo_Correspondences
```

Where: Strict algebraic constraint guaranteeing epipolar line intersection across rectified lenses.

Equation (50.INV.2): SE(3) Group Exponential Invariance

```
Exp(xi_hat) in SE(3), xi_hat = [omega_hat, v; 0, 0]
```

Where: Exact closed-form Rodrigues formulation for rigid body transform integration.

Safety Interlocks, FDIR & Operational Bounds — Part 50

Autonomous failure mitigation for stereo photogrammetry and VIO.

- **Prototype Safeguard:** PROTOTYPE_LOCK = True is held compile-time; navigation_ready evaluates strictly to False.
- **Disparity Discontinuity Sentinel:** If valid disparity pixel count drops below 45% in forward arc, rover velocity automatically decelerates to 2 cm/s.
- **IMU Accelerometer Saturation Interlock:** If IMU shock exceeds 4.5 g, visual feature tracking is quarantined and mechanical brake actuates.

Appendix & Standard Conformance Record — Part 50

Standards conformance for spatial measurement and photogrammetry.

Standard	Description	Status
ISO 10303-242	Industrial Automation Systems — Photogrammetric Geometry	Compliant
ANSI/ASPRS 2014	Positional Accuracy Standards for Digital Geospatial Data	Exceeded
IEEE 1588-2019	Precision Clock Synchronization Protocol for Sensor Triggers	Compliant